MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCAQC200 – Nutrition and feeding of Fishes

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer all 10
Each question carries 1 marks

1. Define digestible energy (DE). [U] [2]

2. Define Specific Growth Rate (SGR) [U] [2]

3. What is linear programming in feed formulation? [U] [3]

4. Why are proteins considered the most important nutrient for fishes? [U][1]

5. What is meant by micronutrients? [U] [1]

6. Why is feed cost important in aquaculture? [U] [4]

7. What is an extruder ? [U] [4]

8. What is lipidosis in fishes? [U] [6]

9. What is feed formulation? [U] [3]

10. What is feeding frequency? [U] [5]

[1x 10=20] Marks

Part B
Short Answer Questions
Answer any 5 Questions out of 8
Each question carries 6 marks

11. Classify vitamins and explain their role in fish nutrition. [U] [1]

12. Describe the different types of feeds used in aquaculture [U] [4]

13. Analyze the significance of growth and feed utilization indices in evaluating fish performance. [An] [2]

14. Demonstrate the application Pearson's square method in feed formulation with a suitable example. [A] [3]

15. Explain the practical use of different feed additives and supplements fish feed. [A] [3]

16. Explain the importance of record keeping in aquaculture feeding management. [U] [5]

17. Discuss feeding rate and feeding frequency at different life stages of fishes. [U] [5]

18. Discuss the relationship between improper feeding practices and nutritional diseases in fish.

[U] [6]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer Any One Question
Each question carries 10 marks

19. Explain the equipment used in feed manufacture and processing. [U] [4]

20. Analyze the relationship between nutrient imbalance and deficiency diseases in fishes.

[An][6]

21. Analyze how dietary energy is partitioned and utilized in fishes under the concept of bioenergetics. [An]
[2]

[10 x 1 = 10] Marks